

Laurent Colbois

POSTDOCTORAL RESEARCHER · VISION-LANGUAGE MODELS FOR FORENSIC BIOMETRICS

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Laurent Colbois



"To create a positive impact in local communities while following and supporting sustainable practices."

Summary

Machine Learning Engineer with a PhD from UNIL/Idiap, transitioning from academic research to industry. Specialized in computer vision and deep learning with a strong focus on reproducible workflows, MLOps practices, and practical implementation. Proven ability to bridge technical and non-technical audiences through outreach to forensic experts and police forces. Seeking to apply expertise in ML engineering to solve real-world problems with meaningful impact.

Work Experience

Idiap Research Institute

Martigny, Switzerland

POSTDOCTORAL RESEARCHER - VISION-LANGUAGE MODELS FOR BIOMETRICS

Apr. 2025 - Mar. 2026

- Benchmarked open-weight Vision-Language Models (Gemma3, Qwen2.5-VL, InternVL3) for explainable face recognition, developing novel metrics to quantify explanation quality and faithfulness
- Implemented reproducible ML pipelines using Hydra, Weights & Biases, and vLLM for efficient VLM inference
- Led project coordination for a forensic data collection platform, managing communication between cross-functional teams (researchers, developers, legal staff) and defining clear technical specifications
- Won first place (15K CHF) at a 9-day startup hackathon by developing a real-time MIDI music generation demo and pitching an AI-based live music improvisation partner to investors and technical audiences
- Submitted research paper to Face and Gesture 2026 on benchmarking VLM explanatory quality

Idiap Research Institute

Martigny, Switzerland

PHD STUDENT - DEEPPAKES, BIOMETRICS, FORENSIC SCIENCE

Jan. 2021 - Mar. 2025

- Developed systems for detection of AI-generated images, with a focus on fake passport photos
- Evaluated the vulnerability of face recognition systems to deepfakes
- Presented outreach talks on AI and deepfakes to the forensic science community and to police forces
- Supervised four master theses on score calibration for forensic face recognition
- Published several research code packages targeting experiments reproducibility

Idiap Research Institute

Martigny, Switzerland

RESEARCH INTERN

Jul. - Dec. 2020

- Created and published a synthetic face recognition dataset using generative AI (GANs)
- Evaluated the applicability of this dataset for benchmarking face recognition systems

Logitech

Lausanne, Switzerland

RESEARCH INTERN

Feb. - Aug. 2019

- Developed an automated audio scene classification model for auto-equalization of a loudspeaker
- Implemented a real-time demo running on a microcontroller

Education

University of Lausanne / Idiap Research Institute

Lausanne, Switzerland

PHD IN FORENSIC SCIENCE (MACHINE LEARNING & BIOMETRICS)

2021 - 2025

- Thesis: Face morphing attacks in the era of deepfakes: risks, detection & source attribution

- Thesis: Trainable filterbanks for end-to-end speech enhancement via time-frequency masking
- Skills: Numerical simulations, Machine learning, Deep learning, Signal processing, High performance computing

- Skills: Applied mathematics, Numerical physics, Programming, Scientific computing

1 Publications

2026	Conference Paper (Under Review) , Benchmarking the Explanatory Quality of Open-Weight Vision-Language Models in Face Recognition	<i>Face and Gesture</i>
2024	Conference Paper , Evaluating the effectiveness of attack-agnostic features for morphing attack detection	<i>IJCB</i>
2023	Conference Paper , Approximating Optimal Morphing Attacks using Template Inversion	<i>IJCB</i>
2022	Conference Paper , On the detection of morphing attacks generated by GANs	<i>BIOSIG</i>
2022	Conference Paper , Are GAN-based morphs threatening face recognition?	<i>ICASSP</i>
2021	Conference Paper , On the use of automatically generated synthetic image datasets for benchmarking face recognition	<i>IJCB</i>

Skills

Domains	Computer vision, Deep learning, Vision-Language Models, Biometrics, Audio processing, Reproducible research, Cross-functional communication
ML & Programming	Python, PyTorch, Huggingface Transformers, vLLM, Scikit-Learn, Pandas, NumPy, C++ (basic)
MLOps & DevOps	Git, Docker, Apptainer/Singularity, Conda, Pip, Pixi, Snakemake, Prefect, Weights & Biases, MLFlow, Hydra, Slurm, Unix/Linux
Languages	French (Mother tongue), English (Professional proficiency), German (B2)